

Treatment Response Across Biofeedback Modalities May Be Mediated by Auditory and Somatosensory Acuity

Purpose: This study examined whether individual differences in auditory-perceptual and oral somatosensory acuity moderate response to different treatment approaches for residual speech sound disorder (RSSD) affecting English /ɹ/. Drawing on a personalized learning framework, we tested whether treatment effectiveness depends on the match between learners' sensory profiles and the sensory information emphasized by intervention.

Method: Participants were 100 children aged 9-15 years who completed a multisite randomized controlled trial comparing visual-acoustic biofeedback, ultrasound biofeedback, and traditional motor-based treatment (MBT). Baseline auditory-perceptual measures included identification, discrimination, and category goodness judgment tasks; somatosensory measures included oral stereognosis and articulatory awareness tasks. Linear regression models tested interactions between sensory acuity and treatment condition in predicting change in perceptually rated accuracy of untreated /ɹ/ words from pre- to post-treatment.

Results: Significant interactions were observed between treatment condition and one auditory measure (discrimination) and one somatosensory measure (oral stereognosis). In the auditory domain, visual-acoustic biofeedback was associated with greater improvement among children with weaker auditory discrimination, whereas ultrasound and MBT showed the opposite pattern. In the somatosensory domain, ultrasound biofeedback and MBT were associated with greater improvement among children with weaker somatosensory acuity, while for visual-acoustic biofeedback the association was reversed.

Conclusions: The observed interactions are consistent with the possibility that treatment effectiveness is influenced by the alignment between learners' sensory needs and the sensory

information highlighted by different interventions. Although replication in larger, balanced samples is needed, the findings provide preliminary support for a personalized approach to treatment planning for RSSD.

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52 **Conflicts of Interest:** There are no conflicts of interest to report.

53 **Keywords:** speech sound disorder; auditory; somatosensory

54 **Running Head:** SENSORY PREDICTORS OF SPEECH TREATMENT RESPONSE

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Individuals with speech sound disorder (SSD) exhibit reduced intelligibility in everyday speech, a difficulty that has been associated with adverse consequences for social engagement and academic and occupational outcomes (e.g., Crowe Hall, 1991; Hitchcock et al., 2015; McCormack et al., 2009). When such deviations persist beyond approximately 8-9 years of age, children may be classified as exhibiting residual speech sound disorder (RSSD; Boyce, 2015). In RSSD, speech deviations typically concentrate on a subset of late-acquired, articulatorily complex phonemes, notably /ɹ/ and /s/. RSSD is widely regarded as difficult to remediate; problems may continue through adolescence and, for an estimated 2% of individuals, into adulthood (Flipsen, 2015). The current study forms part of a larger research initiative aimed at expanding effective treatment options for children and adolescents with RSSD affecting /ɹ/, which has been described as the most frequent residual distortion (Farquharson et al., 2026; Ruscello, 1995). Specifically, the data analyzed here were collected as part of the project C-RESULTS RCT (Correcting Residual Errors with Spectral, ULtrasound, Traditional Speech therapy Randomized Controlled Trial), which measured the efficacy of treatment with and without biofeedback for RSSD affecting /ɹ/ (McAllister et al., 2020).

Biofeedback treatment for RSSD affecting /ɹ/

C-RESULTS RCT was informed by a body of prior research offering evidence that biofeedback-enhanced interventions can help resolve /ɹ/ distortions in children with RSSD (Hitchcock et al., 2017, 2023; McAllister Byun, 2017; McAllister Byun et al., 2014; McAllister Byun & Campbell, 2016; McAllister Byun & Hitchcock, 2012; Preston et al., 2013, 2014, 2018, 2019). In biofeedback, instrumentation is used to generate a real-time representation of aspects of speech that are ordinarily outside conscious awareness (Volin, 1998). A model representing accurate production is displayed simultaneously so the participant can actively modify their

production to better align with the model. Multiple technologies can support biofeedback, including electropalatography (e.g., Hitchcock et al., 2017) and electromagnetic articulography (Vick et al., 2017). C-RESULTS RCT focused on ultrasound biofeedback and visual-acoustic biofeedback. With ultrasound, a probe is positioned beneath the speaker's chin to generate a dynamic image of tongue configuration (Preston et al., 2017). Visual-acoustic biofeedback uses a microphone and software to generate a real-time display of characteristics of the acoustic signal, such as the resonant frequencies of the vocal tract, or formants (Hitchcock et al., 2023). In C-RESULTS RCT, the Sona-Match module of the KayPentax Computerized Speech Lab (CSL) was used to provide a real-time Linear Predictive Coding (LPC) spectrum, where formants appear as peaks in a wave-like shape.

The primary outcome of the C-RESULTS RCT was reported in McAllister et al. (2026). 108 children aged 9–15 with RSSD affecting /ɪ/ were randomized to receive visual biofeedback (subdivided into ultrasound and visual-acoustic types) or motor-based treatment (MBT) representing best practices in traditional intervention. Practice was structured to be as similar as possible across treatment conditions with respect to the number of trials elicited, frequency and type of verbal feedback, and so on. The research question of primary interest (as preregistered in McAllister et al., 2020) looked for between-condition differences in progress over the first three sessions of treatment, termed the acquisition phase. The focus on this phase was grounded in the principles of motor learning (e.g., Maas et al., 2008), which hold that detailed qualitative feedback such as biofeedback can be expected to have its greatest impact in the initial establishment of a new speech-motor plan. Improvement over the first three sessions was indexed by an acoustic measure, the distance between the second and third formants (F3–F2 distance), which is smaller for perceptually accurate /ɪ/ (Campbell et al., 2018; Dugan et al.,

2019). Linear mixed-effects models showed a significant interaction between treatment condition and time: both groups improved, but gains were 2.4 times faster with biofeedback than in the MBT condition. No significant difference was observed between ultrasound and visual-acoustic types of biofeedback. Thus, C-RESULTS RCT provided the first high-quality evidence that biofeedback accelerates acquisition of /ɪ/ in children with RSSD.

The secondary outcome of C-RESULTS RCT was reported in Preston et al. (2026). That study analyzed the 102 children who completed the full course of treatment, which included 16 sessions of generalization-focused treatment following the initial acquisition phase. Children's accuracy in producing /ɪ/ in untreated words elicited at pre- and post-treatment time points was judged by lay listeners blinded to participant characteristics, treatment assignment, and time point. Mixed-effects models showed a statistically significant interaction between time and treatment: both groups improved, but generalization to untreated words was higher for biofeedback than the MBT condition. In a sub-analysis of biofeedback modalities, ultrasound yielded greater generalization than visual-acoustic biofeedback, though this effect was most pronounced at one treatment site.

Personalized learning in treatment for RSSD

In both McAllister et al. (2026) and Preston et al. (2026), no specific hypothesis was advanced regarding the relative efficacy of different types of biofeedback. The rationale was that ultrasound and visual-acoustic biofeedback may support speech learning through distinct mechanisms, with the result that different types of biofeedback may be more effective for different types of learners. Here we draw on second-language (L2) research in what has been described as a personalized learning framework (Perrachione et al., 2011; Wong et al., 2017), which aims to optimize outcomes by measuring learners' baseline profiles and identifying

predictors of success in different training paradigms. Perrachione et al. (2011) illustrated this approach in a study of English speakers learning lexical tones: participants with higher baseline performance on a Pitch Contour Perception Test (PCPT) benefited more from high-variability than low-variability training, but the reverse was true for participants who performed poorly on the PCPT. This work suggested that improved outcomes may be achieved when learners are matched to training paradigms aligned with their strengths and needs.

To identify relevant predictors of response to biofeedback training, we look to current neurocomputational models of speech-motor control such as DIVA (Guenther, 2016) and FACTS (Parrell et al., 2019), which hold that the targets of speech are regions in multidimensional sensory space. These sensory targets span both the auditory domain (i.e., the sound of speech) and the somatosensory domain (i.e., the sense of touch and position of the articulators during speech). There is strong evidence that auditory and somatosensory systems actively shape both the development and ongoing control of speech. Individual differences in auditory-perceptual and somatosensory acuity have been linked to differences in the magnitude of separation between phonemic categories in acoustic and/or articulatory space (e.g., Ghosh et al., 2010; Newman, 2003; Perkell, Guenther, et al., 2004; Perkell, Matthies, et al., 2004; Villacorta et al., 2007). In the clinical literature, it has been documented that children with SSD show reduced average auditory acuity compared to peers with typical speech production, especially for sounds they misarticulate in production (e.g., Cabbage & Hitchcock, 2022; Cialdella et al., 2021; Hearnshaw et al., 2019). The somatosensory domain is less thoroughly documented, but multiple studies have found group differences in oral somatosensory acuity between adolescents with RSSD and age-matched peers with typical speech (Fucci, 1972; Fucci & Robertson, 1971; McNutt, 1977). Other research has suggested that oral somatosensory acuity

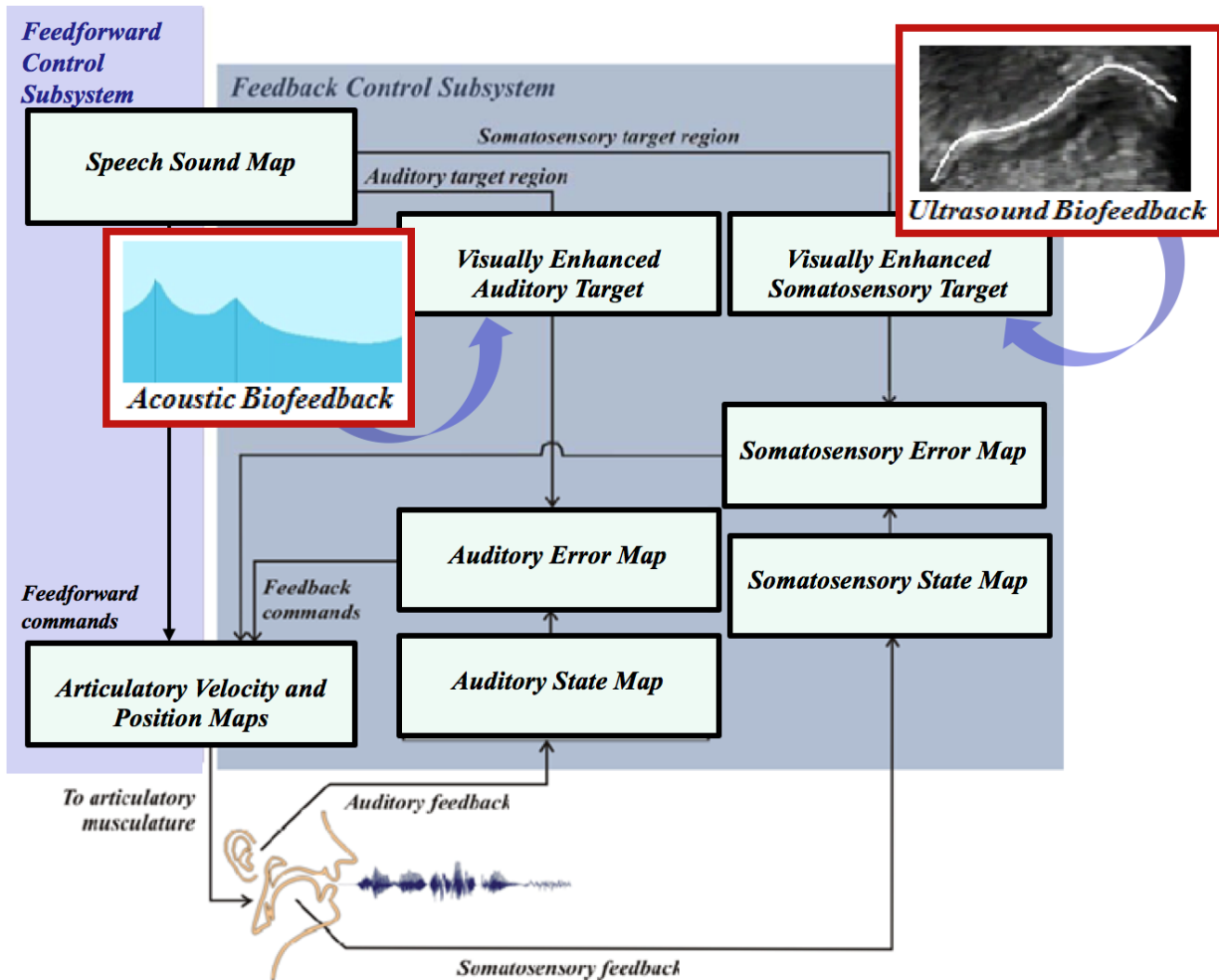
may be linked to the ability to form complex tongue shapes for /ɪ/ in children with RSSD (Kabakoff et al., 2022).

However, it cannot be assumed that every child with atypical speech production will exhibit atypical perception in auditory and somatosensory domains. First, while multiple studies have documented differences in sensory function between children with typical speech and children with SSD, there is significant heterogeneity underlying these group average differences, with some individuals with SSD exhibiting sensory performance indistinguishable from that of peers with typical speech (Cabbage & Hitchcock, 2022; Cialdella et al., 2021; Hearnshaw et al., 2019). Second, auditory and somatosensory acuity have been shown to be independent of one another (Ghosh et al., 2010). Previous research has argued that speakers may respond more strongly to feedback in one domain than the other, sometimes termed a sensory preference (Lametti et al., 2012), or show higher acuity in one domain or the other, termed a sensory tradeoff. For instance, in a study of 100 first-grade children, Madison and Fucci (1971) found a significant negative correlation between scores on oral form discrimination and auditory discrimination tasks, such that children with higher somatosensory acuity tended to perform worse in the auditory domain. However, in their study of children and adolescents with typical speech, Ayala et al. (2023) found no evidence of a tradeoff between oral somatosensory acuity and auditory-perceptual acuity. Still, it is reasonable to maintain that some children show an asymmetry in sensory response between auditory and somatosensory domains, and we posit that such asymmetries could be exploited to inform the selection of a biofeedback treatment modality in a personalized learning framework.

In previous work, we have argued that different types of biofeedback tend to enhance different aspects of the sensory experience of producing speech (Ayala et al., 2023; Benway et

al., 2021; Li et al., 2019). Visual-acoustic biofeedback provides a formant frequency target and a real-time display representing the acoustic signal of speech. These correspond closely with auditory targets and feedback but convey only indirect cues about articulator placement. By contrast, ultrasound imaging provides information about the shape and position of the tongue that are habitually experienced through somatosensory channels; it does not map transparently onto the information ordinarily processed through auditory channels. We therefore hypothesize that treatment effectiveness will depend on the interaction between learners' sensory profiles and the sensory information emphasized by intervention. Specifically, learners who show relative weakness in a given sensory domain may derive the greatest benefit from treatment approaches that enhance that sensory domain; we will call these *congruent* treatment approaches. Thus, children with weak auditory perception should benefit most from visual–acoustic biofeedback, whereas children with weak somatosensory perception should derive more benefit from ultrasound feedback. Conversely, approaches that do not emphasize a given domain may rely more heavily on learners' preexisting strength in that domain; we will term these *orthogonal* treatment approaches. In this paper, we treat visual-acoustic biofeedback as the congruent treatment for the auditory-perceptual domain and ultrasound biofeedback as the congruent treatment for the oral somatosensory domain. The hypothesized relationship between biofeedback modalities and sensory domains is schematized in Figure 1.

Figure 1. Modified schematic of the DIVA model (adapted from Guenther, 2016) illustrating how different biofeedback modalities are hypothesized to interface with auditory and somatosensory domains.



This study conducts a re-analysis of data from the C-RESULTS RCT to evaluate whether sensory measures are in fact predictive of differential response to distinct types of treatment. Specifically, we examined whether auditory and/or somatosensory acuity interact with treatment condition (visual-acoustic biofeedback, ultrasound biofeedback, or MBT) in predicting the magnitude of change over the course of intervention. Such a finding would support the idea that sensory profiles can be used to personalize the choice of a biofeedback treatment modality as a means to optimize treatment outcomes for children with RSSD.

Methods

Participants

This study was conducted under protocol #18-10-393-524, approved by the Biomedical Research Association of New York. All participants provided written informed assent and a parent or guardian provided written consent for their child's participation.

A total of 100 participants completed the full course of treatment in C-RESULTS RCT and were included in the present analysis.¹ The included participants averaged 10.9 years of age (SD = 1.6 years); 35 were female and 65 were male, a distribution aligned with reports of higher SSD prevalence in males (Shriberg, 2009). Of the 100 included children, 41 were assigned to the MBT condition and 59 to the biofeedback condition. More children were allocated to receive biofeedback than MBT (3:2 ratio) to increase power for comparisons between the two biofeedback modalities represented. Within the biofeedback condition, 30 participants received visual-acoustic feedback and 29 received ultrasound feedback.

Eligibility was determined using a pre-screening phone call followed by an in-person evaluation session. For inclusion, participants were required to speak English as a dominant or equally dominant language and to have started learning English by age 3. They were required to have no history of sensorineural hearing loss, major neurobehavioral disorder, or developmental disability, per parent report. Participants also had to pass a pure-tone hearing screening at 20 dB HL and a brief screening of oral structure/function, and they were required to score within normal limits on the Clinical Evaluation of Language Fundamentals-5 (either the screening measure or the Core Language Index). To ensure comparable severity, participants could score no more than 30% correct on a standard word probe of /ɪ/ across multiple phonetic contexts. Childhood apraxia of speech (CAS) was screened with the Syllable Repetition Task (Shriberg et

¹ 108 participants were originally enrolled, but six dropped out prior to the completion of all treatment and are not included here. In addition, two were excluded from analysis due to poor audio quality of probe measures (n = 1) or total dosage falling below a minimum threshold (n = 1).

al., 2009) and the Multisyllabic Word Task of the LinguiSystems Articulation Test (Bowers & Huisinigh, 2011); children exceeding CAS cutoffs on both tasks were excluded. If only one CAS screening measure yielded an elevated score, a maximum-performance task (Thoonen et al., 1996, 1999) was administered as a tie-breaking measure; participants had to score in the typical range on two of the three tasks for inclusion.

Treatment: Elements Shared Across Conditions

Phase 1 (Acquisition) consisted of three 90-minute sessions within two weeks, targeting /ɪ/ at the syllable level across phonetic contexts (syllabic /ɜ/, postvocalic /ɪ/ with front and back vowels, and onset /ɪ/ with front and back vowels). In the first session, participants received instructions consistent with their randomly assigned intervention condition, including an orientation to biofeedback as appropriate. During pre-practice, clinicians modeled each syllable and gave feedback specific to the participant's condition until each target was produced correctly three times or 44 minutes elapsed. Participants were then engaged in structured practice consisting of 150–200 trials grouped into blocks of ten. Clinicians provided Knowledge of Performance (KP) feedback that described the desired articulatory placement (e.g., “Move the back of your tongue farther back”); for biofeedback conditions, feedback also referred to the visual display (e.g., “Align the third peak with the target”; “Try to move your tongue back to the marker”). Participants practiced the same syllable targets with the same frequency of feedback across all three Acquisition sessions.

Phase 2 (Generalization) involved sixteen biweekly one-hour sessions. Each began with 10 minutes of pre-practice, followed by up to 250 structured trials on /ɪ/ in syllables, words, phrases, or sentences. In biofeedback conditions, the biofeedback display was always made

available during pre-practice, but it was faded during structured practice if participants demonstrated a high level of accuracy, as described below.

Task difficulty was automatically adjusted after each block of ten trials using custom software, the Challenge Point Program (CPP; McAllister et al., 2021). The CPP aims to keep learners at an optimal level of practice (Guadagnoli & Lee, 2004) by increasing difficulty if clinician ratings indicate accuracy of 80% or greater and decreasing difficulty if accuracy falls to 50% or below. Practice difficulty was manipulated by adjusting stimulus complexity, feedback frequency, and mode of elicitation. In biofeedback conditions, the frequency of visual feedback decreased from 80% to 0% of trials as participants advanced through levels.

Treatment: Condition-Specific Elements

Motor-Based Treatment (MBT). In the MBT condition, participants received traditional cueing without real-time visual feedback in both Phase 1 and Phase 2. To encourage accurate production, clinicians provided auditory models, verbal descriptions of articulator placement, and visual aids that could be static or animated but were not contingent on the learner's speech. Cues were grounded in established articulatory knowledge of /ɪ/ (Preston, Benway et al., 2020). A standardized list of possible cues for articulator placement was used across sites to promote consistency.

Ultrasound Biofeedback. This condition incorporated all MBT cueing plus real-time visualization of tongue shape and movement (Preston et al., 2017). Ultrasound images were displayed using an EchoBlaster or MicrUS system (Telemed Medical Systems) or an Acuson X300 (Siemens Corporation), viewed via Echo Wave or AverMedia software. Sheet protectors or digital tools (e.g., EpicPen) could optionally be used to annotate the image. A chest-mounted stabilization plate (Hitchcock & Reda, 2020) secured the probe while allowing head movement.

271 Either coronal or sagittal imaging orientations were available at the clinician's discretion.
272 Clinicians emphasized that multiple tongue shapes could yield a correct /ɪ/ and encouraged
273 participants to experiment to find their best fit. To foster independence, the visual display was
274 shown for the first eight of ten trials per block and hidden for the last two, with further
275 reductions in visual feedback frequency as performance improved in Phase 2.

276 **Visual-Acoustic Biofeedback.** Participants in this condition received MBT cueing plus a
277 real-time Linear Predictive Coding (LPC) display generated by the Sona-Match module of the
278 PENTAX Medical Computerized Speech Lab (CSL, model 4500B). Participants were informed
279 that the peaks in the LPC spectrum, which represent formant frequencies, have different
280 positions for different sounds; they were taught that an accurate /ɪ/ should have a low frequency
281 of the third peak and relatively little space between the second and third peaks (F3–F2 distance).
282 Each participant was assigned an LPC template from a child or adolescent with similar formants
283 for sounds other than /ɪ/. Clinicians encouraged participants to adjust their articulator placement
284 to try to achieve a better match with their acoustic template. As in the ultrasound condition,
285 visual feedback was available for the first eight trials in each block and withheld for the last two
286 to promote attention to internal cues, with further reductions in visual feedback frequency as
287 accuracy increased in Phase 2.

288 **Outcome Measure**

289 Participants' learning was assessed with standard generalization probes in which each
290 child read 50 untreated /ɪ/ words representing a range of phonetic contexts. This study analyzes
291 the probe from the initial evaluation conducted before the start of any treatment and the post-
292 treatment evaluation conducted ~1 week after the final session. (These are the same data reported
293 in Preston et al. (2026), but with a different analysis.) Recordings at all sites used a head-

mounted AKG C520 mic routed to a Marantz solid-state recorder via a Behringer UMC404HD interface. Audio was captured at 44.1 kHz/16-bit as lossless PCM WAV.

Each probe was segmented into 50 word-level recordings in Praat (Boersma & Weenink, 2024); these were pooled across speakers and time points and randomized for rating. Ratings were obtained from everyday listeners recruited through Amazon’s Mechanical Turk platform. For inclusion, raters had to have a U.S.-based IP address and self-report as native speakers of American English with no history of hearing, speech, or language difficulty. Participants also had to pass a 100-item qualification test based on expert listeners’ judgments of words containing /ɪ/ (McAllister Byun et al., 2015). Raters saw the orthographic representation of the target word and were instructed to score each word as “correct” or “incorrect” based on the accuracy of the /ɪ/ sound, ignoring any deviations elsewhere in the word. Ratings were collected until at least nine unique listeners had judged each token; the proportion of “correct” ratings served as our primary measure of participants’ /ɪ/ production accuracy.

Sensory Predictors

Baseline measures were collected across two visits (1–2 hours each) on separate days prior to the start of the intervention. In Session 1, participants completed all eligibility assessments, including hearing and oral mechanism screenings and standardized test administration. Participants then completed a battery of auditory and somatosensory measures detailed in Ayala et al. (2023) and summarized below. Listening tasks were presented over Sennheiser HD 280 PRO headphones (frequency response: 8 Hz–25 kHz) in a quiet room or sound-shielded booth. Because audio was delivered on different devices across three sites, we first completed a calibration procedure to equate output level across systems (see Ayala et al., 2023 for additional detail). Auditory tasks comprised: (1) identification along a synthesized

“rake”–“wake” continuum; (2) AXB discrimination of pairs from the same continuum; and (3) category-goodness judgments of /ɹ/ in naturally produced words from speakers of varying ages and sexes, with and without RSSD. Somatosensory tasks (initially developed in Gritsyk et al., 2021) included: (1) oral stereognosis, in which participants identified raised letters on plastic strips using the tongue; and (2) an articulatory-awareness task assessing explicit judgments of articulator position. A third task—compensation to a bite block perturbation during auditory masking noise—was excluded due to excessive data loss associated with COVID-19 sanitary precautions. All materials (stimuli, protocols, scripts) are available on the Open Science Framework at <https://osf.io/bkasm/>.

Auditory Measures.

Identification. Isolated productions of the words “rake” and “wake” were obtained from a 13-year-old female speaker. Two tokens similar in duration and pitch contour were selected as endpoints for a 240-step continuum generated with STRAIGHT synthesis (Kawahara et al., 2013), from which ten tokens, roughly evenly spaced in F3, were selected for the identification task. Participants were instructed to identify each token by clicking “rake” or “wake.” After a brief training using the endpoints of the continuum, they heard each of the ten steps eight times in a randomized order (80 trials total). The proportion of “rake” responses to each step of the continuum was plotted and fitted with a logistic function through maximum likelihood estimation. The width of the boundary region from the 25th to the 75th percentile of probability was used as a measure of response consistency; narrower boundaries are considered indicative of more mature perception (McAllister Byun & Tiede, 2014). This task is thought to tap a categorical mode of perception (Cheng et al., 2021; Pisoni, 1973).

Discrimination. The same 240-step continuum from “rake” to “wake” was used in an AXB discrimination task implemented in MATLAB. In each trial, participants heard three tokens separated by 500 ms; they were instructed to select which of the two flanking stimuli differed from the middle stimulus. Stimulus pairs were centered on each participant’s /ɹ/-/w/ boundary as determined by the identification task. The separation between the A and B tokens was adaptively adjusted using an automated staircase (initially decreasing by eight steps after a correct response and increasing by four steps after an incorrect response, then halving step sizes until a two-down, one-up configuration was reached). The task ended after 12 reversals or 80 trials. Just noticeable difference (JND) was computed as the mean step size across the final four reversals, with smaller JNDs reflecting more acute perception. To sustain motivation, the task was presented in a child-friendly, gamified format. Compared to the identification task, the discrimination task is thought to tap a lower-level auditory mode of perception, although the categorical mode is likely still engaged to some extent (Cheng et al., 2021; Pisoni, 1973).

Category goodness judgment. Participants’ ability to judge the accuracy of others’ /ɹ/ productions was assessed in a 100-item task administered in Praat (Boersma & Weenink, 2024). Stimuli were naturally produced words from 52 speakers (adults and children with and without RSSD). Each item contained a single /ɹ/ sound in onset, nucleus, or postvocalic position. Gold-standard ratings were established by consensus across at least four expert listeners; half of the items had a consensus rating of “correct” and half of “incorrect.” On each trial, participants heard the word and were instructed to “choose whether the /r/ sound is right or wrong.” No training phase was provided for this relatively straightforward task.

Somatosensory Measures.

Oral stereognosis. As an index of oral tactile acuity, participants identified raised letter forms using their tongue tip. The stimuli and protocol followed Steele et al. (2014). Stimuli were single sans serif letters embossed on Teflon strips with seven letter heights ranging from 2.5 to 8 mm. Participants were given each strip face-down and were instructed to search the embossed portion with the tongue tip, then name the letter. The stimuli were presented in an adaptive staircase beginning at the 5 mm size, with letter size decreasing after a correct response and increasing after an incorrect response. Stimuli were presented until eight reversals occurred. Spatial resolution acuity was calculated as the mean letter size (mm) across correct responses through the eighth reversal.

Articulatory awareness. This task, originally developed in Gritsyk et al., (2021) assessed participants' proprioceptive awareness of speech movements. PsychoPy 2 (Peirce et al., 2019) was used to present written and audio prompts eliciting specific speech sounds (e.g., "say *ee* as in *heat*"). Participants were prompted to repeat each target three times while attending to the position of their tongue, then answer a question about tongue position. Block 1 asked whether a consonant (e.g., /t/) used the front or back of the tongue. Blocks 2–4 asked participants to produce two vowels in isolation three times (e.g., /u/–/i/, /u/–/i/, /u/–/i/), then judge relative tongue position (which was farther back/higher/lower). Responses were given verbally by the participant and entered into the software by the examiner. The primary outcome was percent correct across 34 items, with higher scores indicating greater proprioceptive awareness.

Analyses

Study data were managed in REDCap (Harris et al., 2009, 2019). A trained team member verified entries and flagged anomalous patterns (e.g., suggestive of random responding) for review and possible exclusion. Because several measures were non-normal on visual inspection,

we used nonparametric methods for summaries and associations. Data cleaning followed a robust rule (Leys et al., 2013) that treats values as outliers if they fall ≥ 3 scaled median absolute deviations from the median within each group (typical or RSSD). The exclusion of corrupted data and outliers resulted in the exclusion of the identification task from $n = 7$ participants, the discrimination task from $n = 9$ participants, the category goodness judgment task from $n = 2$ participants, and the articulatory awareness task from $n = 1$ participant. In addition, the stereognosis task represented a special case where data were missing from a large number of participants due to sanitary precautions during the COVID-19 pandemic; this resulted in missing data from $n = 54$ participants. Given that the sources of missingness appear to be largely random with respect to our treatment groups, we anticipate that the exclusion of these data will not bias our results.

Analyses were carried out in R (R Core Team, 2023), with the tidyverse family of packages used for data wrangling (Wickham et al., 2019). Linear regression models were used to examine treatment progress as predicted by randomization group and sensory acuity measures, as well as two-way interactions between randomization group and each sensory measure. Separate models were used for auditory and somatosensory predictors. The outcome variable in the results reported below was change in average perceptually rated accuracy of /ɪ/ probe words from the pre- to the post-treatment time point. We also examined similar models that included accuracy at the pre-treatment time point as a controlled covariate, since previous research has found baseline accuracy to be predictive of response to intervention (Preston et al., 2026). These two modeling options were substantively similar with respect to the significance and direction of effects; complete results of both the original and the alternative models can be found in Supplementary Table A.

The effect that was of primary experimental interest was the interaction between baseline sensory acuity and randomization group. A few modeling choices were necessary for the predictor variables representing sensory acuity. First, it was necessary to determine whether sensory measures should be pooled within a domain (e.g., using the average across all three auditory measures as a single predictor variable) or if the individual tasks should stand as separate predictors. We found only weak correlations between scores on the three auditory measures (maximum correlation of $r = .27$, observed for identification and discrimination scores) and therefore opted to represent them separately in the auditory model. Likewise, the two somatosensory measures were correlated with $r = .12$; they are represented separately in the somatosensory model. Complete pairwise correlations are represented in Supplementary Table B.

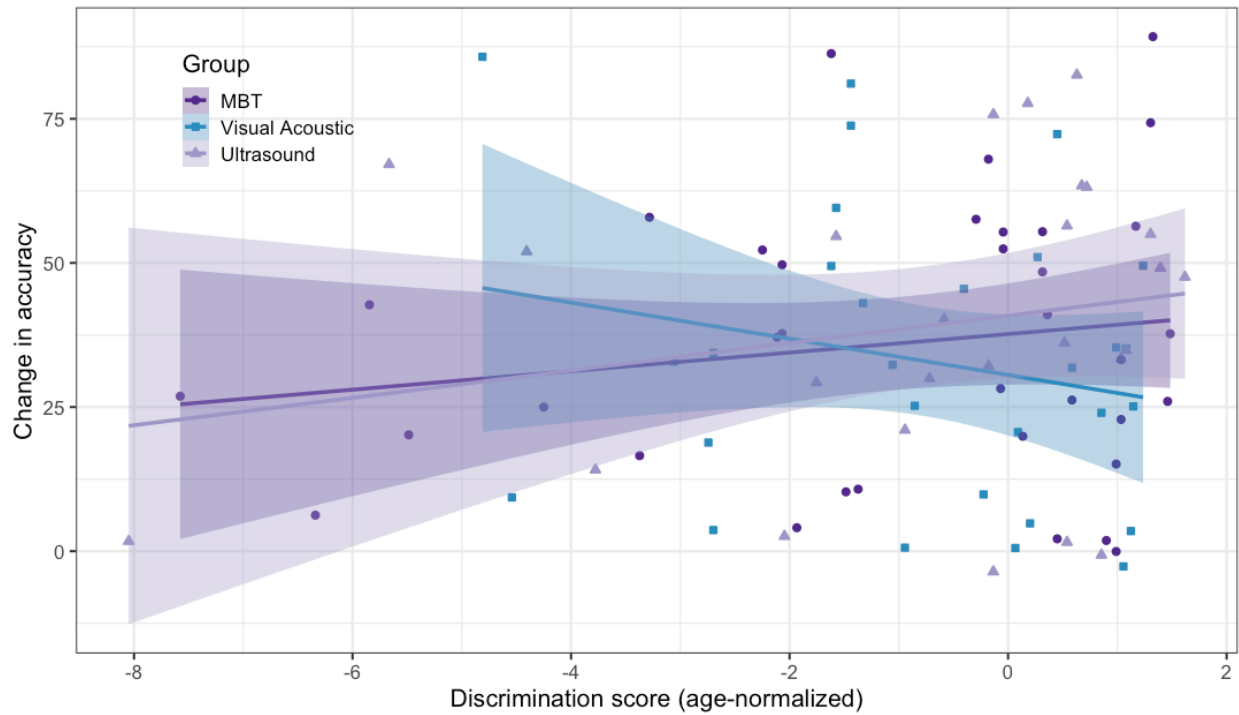
Second, our previous study collecting reference data on the auditory and somatosensory tasks (Ayala et al., 2023) indicated that some measures were significantly correlated with age. Ayala et al. (2023) used model-based clustering to identify empirically derived age bands, yielding a younger group (9–12.5 years) and an older group (>12.5 years). To adjust for the potential effects of age on sensory performance in the present study, scores on each task were normalized to reflect each participant's degree of deviation from the central tendency of scores from typical peers within the appropriate age band from Ayala et al. (2023). Because distributions were non-normal, deviations were expressed in median absolute deviations (MAD) above or below the typical group median. Finally, since measures differed in whether higher or lower scores represented more acute sensory function, three measures (identification, discrimination, and stereognosis) were reverse-coded so that higher values uniformly reflected better performance. In the rest of this paper, we will use “identification score,” “discrimination

score,” “category goodness judgment score,” “stereognosis score,” and “articulatory awareness score” to refer to the age-normalized and (when needed) reverse-coded measures of performance that were used in the analyses reported below.

Results

In the model examining auditory predictors ($n = 85$), there was a positive direction of association between discrimination score and change in accuracy in the ultrasound biofeedback treatment condition, although it fell just short of the conventional threshold for statistical significance ($\beta = 4.45$, $SE = 2.27$, $p = 0.054$). By contrast, the slope of the association for visual-acoustic biofeedback was negative, such that children with lower discrimination scores tended to exhibit a larger degree of change over the course of treatment than children with higher discrimination scores; see Figure 2. Although this slope was not significant on its own, the interaction between discrimination score and treatment group was statistically significant for the visual-acoustic versus ultrasound comparison ($\beta = -9.49$, $SE = 4.03$, $p = 0.021$). MBT, like ultrasound biofeedback, showed a positive direction of association between discrimination score and change over the course of treatment, but this association did not reach or approach significance. There were no significant effects or interactions for the other auditory measures (identification and category goodness judgment); see Supplementary Table A for full results.

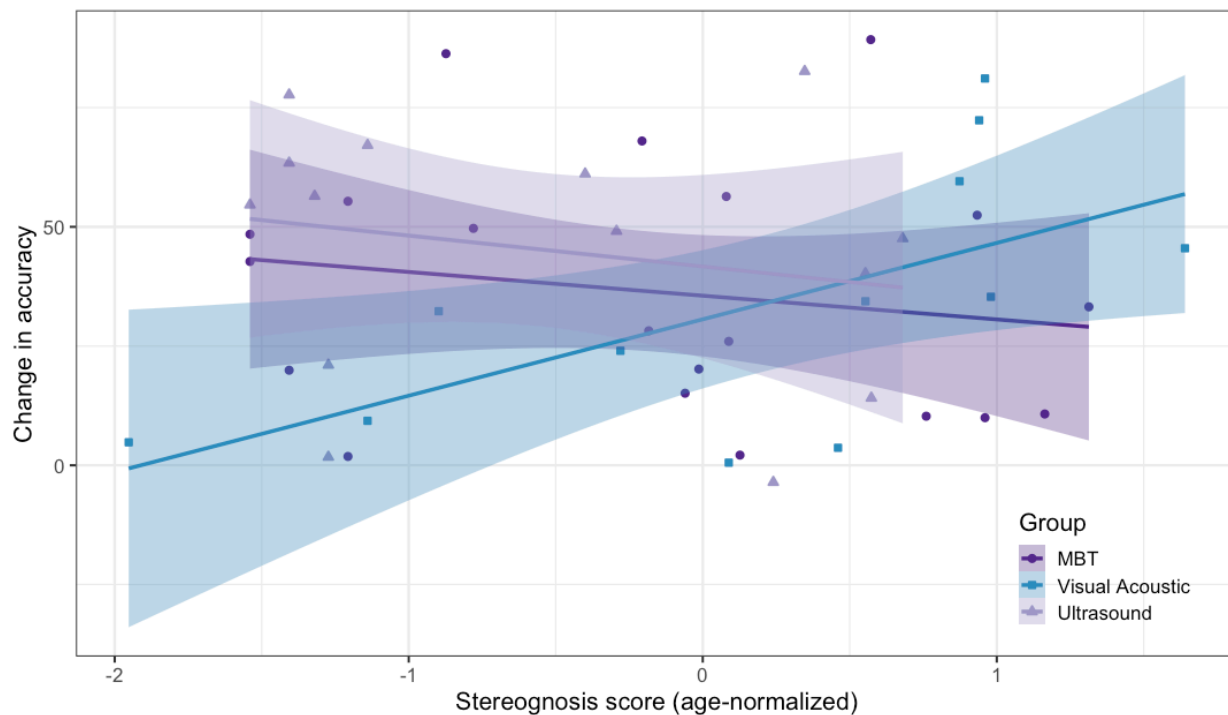
Figure 2. Relationship between discrimination score (age-normalized JND, reverse-coded such that a higher score indicates more acute perception) and change in perceptually rated accuracy of /ɹ/ production from pre to post treatment. JND = Just Noticeable Difference.



In the model examining somatosensory predictors ($n = 46$), there was a positive association between stereognosis score and change in accuracy in the visual-acoustic biofeedback treatment condition ($\beta = 18.91$, $SE = 9.06$, $p = 0.044$), but not in the other conditions. There were significant interactions between age-normalized stereognosis score and treatment group (visual-acoustic versus MBT: $\beta = 24.72$, $SE = 11.18$, $p = 0.033$; visual-acoustic versus ultrasound: $\beta = 25.53$, $SE = 12.33$, $p = 0.045$). Visual inspection of Figure 3 reveals that MBT and ultrasound biofeedback showed a slightly negative relationship between stereognosis score and change over the course of treatment, such that children with lower stereognosis scores tended to exhibit a larger treatment response than children with higher scores. This contrasted with the positive direction of association between stereognosis score and response to treatment in the visual-acoustic biofeedback condition.

Figure 3. Relationship between stereognosis score (age-normalized mean letter size, reverse-coded such that a higher score indicates more acute perception) and change in perceptually rated

466 accuracy of /ɪ/ production from pre to post treatment.



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Discussion

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This study asked whether baseline measures of auditory and somatosensory acuity would be predictive of response to different types of treatment for children with RSSD affecting /ɪ/. As part of a larger study, children were randomly assigned to receive treatment featuring visual-acoustic biofeedback, ultrasound biofeedback, or traditional motor-based treatment (MBT) informed by the principles of motor learning. Our particular objective, aligned with research in the personalized learning framework, was to understand whether auditory and somatosensory measures would differentially predict response to biofeedback modalities that highlight distinct aspects of the ordinary sensory experience of producing speech. Specifically, we hypothesized that visual-acoustic biofeedback, which adds visual support to auditory-acoustic targets and feedback, would offer greater relative benefit for learners with weak auditory acuity than those

whose auditory perception was already robust. We thus identified visual-acoustic biofeedback as the congruent treatment condition for the auditory domain; ultrasound biofeedback and MBT were characterized as orthogonal treatment conditions. Similarly, we predicted that ultrasound biofeedback would offer greater relative benefit to learners with weaker somatosensory acuity, since ultrasound provides a visual display of information about articulator placement that is ordinarily received through somatosensory channels. Therefore, ultrasound biofeedback was identified as the congruent treatment condition for the somatosensory domain, and the other two treatment conditions were characterized as orthogonal.

Our findings were broadly aligned with this hypothesis. In the auditory domain, there was a significant interaction between treatment condition and sensory acuity as measured with the discrimination task. Among participants who received visual-acoustic biofeedback, the direction of the trend between discrimination score and change over the course of treatment was negative, indicating that weaker performance on the perceptual measure was associated with greater response to treatment, as predicted for the congruent condition. Ultrasound biofeedback and MBT showed slopes in the opposite direction, such that stronger auditory discrimination tended to be associated with greater improvement in those conditions, as predicted for orthogonal conditions. A positive association between auditory-perceptual acuity and response to ultrasound biofeedback intervention was also observed in a previous study, although in that case the associated task was identification rather than discrimination (Preston et al., 2020). In the somatosensory domain, there was a significant interaction between treatment condition and stereognosis score. Ultrasound biofeedback and MBT showed slopes in the negative direction, indicating that weaker performance on the somatosensory measure tended to be associated with

greater response to treatment. By contrast, in the visual-acoustic biofeedback condition there was a statistically significant positive association between stereognosis score and treatment response.

These findings are compatible with the idea that biofeedback may be most effective when it fills an area of need for a learner, such as visual enhancement of the auditory signal for learners with weak auditory perception. For treatment approaches that do not enhance a given sensory domain, performance may depend more heavily on learners' preexisting strength in that domain. This could explain why the association between sensory acuity and treatment response generally showed a positive trend when participants received a type of feedback that did not target that domain (e.g., auditory acuity for participants who received ultrasound biofeedback treatment).

Differences across sensory measures

A few questions are left unanswered by the present findings. First, it is not clear why the interaction between sensory acuity and treatment condition was observed for only one sensory measure in each domain; comparable interactions were not present for the identification and category goodness judgment tasks in the auditory domain, or for the articulatory awareness task in the somatosensory domain.

Our choice to represent the different sensory measures separately in our analyses, instead of averaging or otherwise pooling across tasks, was motivated by the observation that correlations between tasks tapping a given domain were generally weak. One possible interpretation is thus that the different measures assess independent facets of each sensory domain. For instance, it has long been argued that speech perception has distinct auditory and categorical modes (Pisoni, 1973). Out of our auditory-perceptual measures, the discrimination task is noteworthy as the most likely to tap the low-level auditory mode rather than the

categorical mode of perception (Cheng et al., 2021). The identification task assesses the ability to consistently assign a categorical label to stimuli with intermediate acoustic properties, and the category goodness task requires more metalinguistic judgments as to what constitutes a correct /ɪ/ sound. Thus, it is possible that visual-acoustic biofeedback is most beneficial for learners who have difficulty hearing the raw difference between their productions and typical /ɪ/ sounds, as compared to learners who have difficulty consistently classifying ambiguous productions or those who struggle to make adult-like judgments of accuracy. A potential explanation, albeit speculative, is that visual-acoustic biofeedback can visually reinforce incremental progress in the direction of the target acoustics for /ɪ/, even when the learner is not yet crossing or even approaching the category boundary. Such feedback could be especially valuable for individuals who have difficulty using the auditory modality to understand which articulatory adjustments are moving them in the direction of a perceptually accurate /ɪ/.

In the somatosensory domain, we did not include a true discrimination task, such as a two-point discrimination threshold. However, it is reasonable to assert that the stereognosis task is the closest counterpart of discrimination available in the somatosensory battery used for this study. The articulatory awareness task has a prominent metalinguistic component that makes it more akin to the category goodness judgment task in the auditory domain. Thus, the same speculation we offered for the case of visual-acoustic biofeedback can be applied here: the measure that most closely evaluates raw ability to discriminate between two sensory experiences is most predictive of response to biofeedback treatment. Future research using true discrimination tasks in the oral somatosensory domain, such as two-point discrimination or vibrotactile thresholds, could be useful to further substantiate this hypothesis.

Differences across treatment conditions

An additional unanswered question pertains to the patterning of the MBT condition compared to the two biofeedback treatment conditions. Our specific hypotheses pertained to the relationship between auditory and oral somatosensory acuity and the two biofeedback modalities, visual-acoustic (considered the congruent condition for the auditory-perceptual domain) and ultrasound (considered the congruent condition for the oral somatosensory domain). We characterized MBT as an orthogonal treatment condition with respect to both domains, giving rise to the prediction that the association between acuity and response to MBT should be positive for both sets of sensory measures. What we observed instead, in both auditory and somatosensory domains, was a significant interaction in which ultrasound biofeedback and MBT behaved similarly and visual-acoustic biofeedback patterned separately. That is, MBT behaved as if it were a congruent treatment condition for the somatosensory domain.

One possibility is that the verbal cues provided during MBT encourage a focus on articulator placement, and the visual display of the tongue provided by ultrasound simply augments this preexisting focus. It is important to keep in mind that articulator placement cues were also provided to learners in the visual-acoustic biofeedback condition. However, previous research has argued that visual-acoustic biofeedback can encourage an external direction of attentional focus, even when articulator placement cues are provided (McAllister Byun et al., 2016). Thus, our leading hypothesis for the differential patterning of visual-acoustic biofeedback compared to the other two treatment types is that they differ in the direction of attentional focus they encourage learners to adopt. Specifically, MBT and ultrasound may encourage a greater degree of attention to articulator placement, while visual-acoustic biofeedback may encourage a more external orientation of attention. This possible difference between biofeedback types in the direction of attentional focus warrants further study, particularly in connection with the

observation that ultrasound and visual-acoustic biofeedback yielded a similar rate of progress in the acquisition phase (McAllister et al., 2026), but ultrasound supported a greater degree of generalization (Preston et al., 2026).

Possibility of sensory tradeoff

As noted previously, some studies have found that speakers may show a sensory preference or tradeoff between auditory and somatosensory domains, responding more strongly to feedback in one domain than the other (Lametti et al., 2012; Madison & Fucci, 1971; but cf. Ayala et al., 2023). It is worthwhile to consider whether the present results provide evidence of a tradeoff between sensory domains, since such a finding would be relevant for a personalized learning approach to biofeedback. Examining the full set of pairwise correlations between measures both across and within domains (see Online Supplement B), there were three correlations with a negative directionality, all involving the association between an auditory measure and a somatosensory measure. However, two of these correlations were virtually zero (identification and stereognosis, $r = -0.01$; identification and articulatory awareness, $r = -0.01$), and the third was also very small and non-significant (category goodness and stereognosis, $r = -.15$).

We conclude that the present results do not provide substantive support for the hypothesis of sensory tradeoff across domains. However, it is important to keep in mind that the tasks used to measure sensory function in the present study are not equivalent across domains. The auditory-perceptual tasks specifically targeted perception of American English /ɪ/, which aligned with the deviations our participants with RSSD exhibited in production. In the somatosensory domain, we were unable to identify analogous tasks specific to /ɪ/, so we evaluated somatosensory acuity in a more general way. Future research that measures auditory and

somatosensory feedback on more equal footing could still shed light on our predictions regarding sensory preference and personalized learning. This could be accomplished using simultaneous perturbation as in Lametti et al. (2012), or with auditory and somatosensory gain scores derived from a single task of adaptation to altered auditory feedback, as in Kearney et al. (2020).

Limitations

A limitation of the present study is the relatively small number of data points representing each of the three treatment conditions, particularly after the exclusion of cases of data loss. Prior to exclusions, there were roughly 30 participants in each biofeedback condition and roughly 40 participants in the MBT condition. While per-condition sample sizes on the order of 30 to 40 are common in behavioral studies, statistical power is typically lower for interaction terms than for main effects; accordingly, the present design is most sensitive to moderate-to-large interaction effects, and smaller interaction effects may not be reliably detected. For the analysis of somatosensory measures, the substantial data loss in the stereognosis task due to COVID-related sanitary precautions resulted in much lower counts in each group: 20 participants in the MBT condition, 14 in the ultrasound biofeedback condition, and 12 in the visual-acoustic biofeedback condition. Under these circumstances, parameter estimates for interaction terms are associated with wide confidence intervals and heightened sensitivity to sampling variability, and statistically significant effects may be disproportionately influenced by a small subset of observations. Accordingly, the interaction observed in the somatosensory domain should be viewed as provisional until it can be replicated in a larger and more balanced sample.

There are limitations to the sensory measures adopted in this study, as discussed in Ayala et al. (2023). Oral somatosensory tasks that involve letter recognition have been questioned due to systematic differences in the perceptual salience of individual letters (e.g., I versus F), as well

as the possibility that literacy skills could influence performance (Appiani et al., 2020; Lee et al., 2022). In addition, the version of the task used here may place demands on visuospatial processing, since letters are presented with their canonical orientation reversed relative to the oral cavity and therefore require mental rotation for accurate identification. In the auditory domain, the “rake”–“wake” identification task has been criticized for a pronounced ceiling effect in older children and adolescents with typical speech: approximately 40% of the typically developing participants in Ayala et al. (2023) achieved the minimum possible boundary width of 0. However, Ayala and colleagues argued that the measure can still be informative for clinical populations, since performance on variants of this task has been found to differ between children with typical speech and those with RSSD (Cialdella et al., 2021) and to predict treatment outcomes in children with RSSD (Benway et al., 2021; Preston et al., 2020).

Future directions

The present findings can be compared with previous research examining sensory predictors of biofeedback treatment response in healthy adults learning to produce sounds from a second language (L2). Lai & Chiu (2025) examined numerous predictors of response to ultrasound biofeedback training of vowels that were unfamiliar for speakers of Mandarin, including an articulatory phonetic awareness task similar to that used here (but adapted to the vowel space of Mandarin), as well as a measure of oral somatosensory acuity that involved compensating for the presence of a bite block under conditions of auditory masking. They found that lower performance on the articulatory phonetic awareness task was associated with larger response to ultrasound biofeedback training. This result is consistent with the predictions advanced here, but contrasts in that we did not find statistically significant effects or interactions involving our articulatory awareness task. However, Lai & Chiu (2025) also found a positive

direction of association between response to ultrasound biofeedback and somatosensory acuity as measured with the bite block task. This runs counter to our prediction that ultrasound biofeedback should differentially benefit learners with weak oral somatosensory function. It also conflicts with the findings of Gritsyk et al. (2021), where performance on a bite block measure of oral somatosensory acuity was negatively associated with response to biofeedback training. However, participants in Gritsyk et al. (2021) received either visual-acoustic biofeedback or ultrasound biofeedback, and the feedback types were not analyzed separately. The present study cannot resolve this discrepancy because no bite block measure was available for analysis. In future research, it will be of interest to see whether the relationship between sensory predictors and response to biofeedback treatment differs between healthy adult L2 learners and clinical populations like children with RSSD.

Our long-term goal in undertaking this line of research was to allow clinicians to characterize child clients in terms of their profiles of sensory acuity. This makes it important to consider factors such as the ease of administration of sensory tasks and the attentional demands they impose on child participants. While the auditory discrimination task was most successful in differentiating between treatment types in the present study, Ayala et al. (2023) commented that the task is especially demanding in terms of sustained attention. In addition, the adaptive component of the task is implemented in MATLAB; unlike the other auditory tasks investigated in this study, it does not currently have a web version that can be deployed in a browser. The stereognosis task poses even greater challenges for translation to the clinical setting, since the need to reuse a set of objects that are placed in the mouth necessitates an in-depth sanitary protocol. In addition, the set of letters we used is not commercially available and was costly to custom manufacture, although the wider availability of 3D printers could potentially overcome

this barrier. Thus, while insights from this study have potential to inform clinical practice in the long term, the potential for clinical uptake in the immediate term is limited. In future research, we will aim to build on the current insights using tasks that can be more readily adopted in real-world clinical settings.

Conclusion

This reanalysis of data from a large, multisite randomized controlled trial provides initial evidence that individual differences in sensory acuity may moderate response to different treatment approaches for RSSD. Specifically, interactions between auditory discrimination and oral somatosensory acuity and treatment condition suggest that visual–acoustic biofeedback may preferentially benefit learners with weaker auditory perception, whereas approaches that emphasize articulatory information may offer greater advantage to learners with weaker oral somatosensory acuity. These findings are compatible with a personalized learning framework in which the effectiveness of treatment can be influenced by the alignment between learner profiles and the sensory information highlighted by intervention. At the same time, several limitations of the study indicate that these results should be interpreted with caution. These include modest per-condition sample sizes for interaction analyses, substantial data loss for one somatosensory measure, and the limited clinical feasibility of the sensory tasks used. Future work should seek to replicate these effects in larger and more balanced samples, develop clinically scalable measures of sensory function, and test whether prospective matching of learners to biofeedback modalities based on sensory profiles can improve treatment outcomes.

Acknowledgements. The research team is grateful for the contributions of the following individuals (ordered alphabetically): Samantha Ayala, Molly Beiting, Nina Benway, Nicole

686 Caballero, Twylah Campbell, Erin Doty, Sarah Granquist, Lynne Henry, Danielle Kealy,
687 Roberta Lazarus, Megan Leece, Megan Matson, Laura Ochs, Cory Pinto, Amy Schwartz,
688 Michelle Swartz, and Stephanie Urlass. The authors thank Siemens Corporation for the loan of
689 an Acuson X300 ultrasound device to New York University.

690

691 **Funding Statement.** This research was funded by NIH grants R01DC017476 and
692 2R01DC017476, Biofeedback-enhanced treatment for speech sound disorder: Randomized
693 controlled trial & delineation of sensorimotor subtypes (2019-2029; PI: McAllister).

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695 **Data Availability Statement.** Complete code and de-identified data to reproduce the analyses
696 reported here can be found at <https://github.com/tara-mcallister/sensory-predictors>.

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949

List of Supplementary Materials

950 **Supplementary Table 1.** Complete results of the linear regression models examining auditory
951 predictors (1.1–1.3) and somatosensory predictors (1.4–1.6). Alternative models including a
952 control for baseline accuracy are reported for both auditory predictors (2.1-2.3) and
953 somatosensory predictors (2.4-2.6).

954 **Supplementary Table 2.** Complete table of pairwise correlations between scores on individual
955 sensory measures.

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